Supplementary appendix: key lemmas for the macroscopic conditional law

Self-contained proofs corresponding to Propositions 8 and 10 of the main text

Min Wu

October 14, 2026

F Key lemmas for the macroscopic conditional law

F.1 Scope, notation, and existing inputs

Throughout we fix $k \geq 3$, work in the Plain model, and place the load in the moderate deviation regime:

$$\frac{d}{M} = \alpha_c + \frac{r_M}{\sqrt{M}}, \qquad r_M \to -\infty, \qquad |r_M| \leq C\sqrt{\log M}. \tag{F.1}$$

Write $t_0 = -g_k r_M \to \infty$, where $g_k = k a_c e^{-x_c}/\sigma_k$; $p_M = \Phi(-t_0)$; $\lambda_M = \binom{d}{2}/\binom{M}{k}$. The macroscopic event is $A = \{\text{the number of edges of the actual 2-core} > \varepsilon_0 M\}$. The sign factor is $\gamma_M = (d\rho_M^2 - 1)/(d-1)$; the all-positive-sign measurement is denoted by X_+, and the measurement under an arbitrary fixed sign composition is denoted by X_s. $\ell(t) = \phi(t)/\Phi(-t)$ denotes the left-tail Mills ratio, namely $A(-t)$ of the main text.

This appendix proves four groups of conclusions, related to the main text as follows.

(1) The macroscopic conditional CDF of the all-positive-sign measurement: $\sup_x |\Pr(X_+ \leq x \mid A) - \Phi_2(x, -t_0; \eta_k)/\Phi(-t_0)| \to 0$ (Proposition F.6).

(2) The conditional moments of the all-positive-sign measurement and the conditional mean of an arbitrary sign composition (Proposition F.7).

(3) The macroscopic conditional CDF and conditional variance of an arbitrary sign composition, namely the first two formulas of Proposition 8 of the main text (Proposition F.8).

(4) The control of the three weighted error terms, namely the weighted-mean formula of Proposition 10 of the main text (Proposition F.13).

We directly use the scalar inputs already established in Appendix C (the proof of Proposition 7 of the main text). For ease of reference, they are collected here.

(C1) There is a good event G_M whose complement has super-polynomially small probability. On G_M, up to the stopping time, all histogram errors are at most $C\vartheta_M$, where $\vartheta_M = \sqrt{M}(\log M)^6$.

(C2) In the configuration model, the initial Doob increments and the dynamic increments assemble into a score martingale S; its increments are at most $C/\sqrt{M}$, the bracket has a deterministic upper bound, and, except for a super-polynomially small exceptional probability, it lies within $1 \pm v_M$, where $v_M = C\{M^{-1/8} + (\log M)^8/\sqrt{M}\}$.

(C3) There is a bandwidth sandwich between the macroscopic event and the score event: $A \triangle \{S \leq -b_M\} \subseteq \mathcal{B}_M \cup \{|S + b_M| \leq C\delta_M\}$, where $\mathcal{B}_M$ is super-polynomially small, the deterministic bandwidth $\delta_M \to 0$ satisfies $b_M \delta_M \to 0$, and $b_M = t_0 + O((1 + r_M^2)/\sqrt{M})$.

(C4) The relative asymptotics of the macroscopic probability: $\Pr(A) = p_M(1 + o(1))$; the same holds with l fixed special edges, and it can be transferred to the Plain model via relative Bonferroni.

"Super-polynomially small" means $o(M^{-b})$ for every fixed $b > 0$. By (F.1), p_M is no smaller than a fixed negative power divided by a polylogarithm, so a super-polynomially small quantity divided by p_M remains super-polynomially small. This is used repeatedly below.

F.2 Choice of the macroscopic cutoff

Proposition 7 of the main text holds for any $0 < \varepsilon_0 < \alpha_c q_c^k$. The two-regime reduction also requires the small-core counts to cover all core sizes below the macroscopic layer, so this appendix fixes a sufficiently small cutoff.

Lemma F.1 (Choice of the cutoff)**.** *Take $\varepsilon_* > 0$ such that the small-core summation of Eq. (64) of Appendix B holds for $r \in \{0, 2\}$ on $3 \le e \le \varepsilon_* M$, and such that $\varepsilon_* < \alpha_c q_c^k/2$. Fix $0 < \varepsilon_0 \le \varepsilon_*$. Let D be the event that a repeated edge pair exists, and let $R = F \setminus (A \cup D)$. Then under (F.1)*

$$\Pr(R) = O(M^{-b_k}) = o(\lambda_M), \qquad \Pr(A \cap D) = o(\lambda_M). \tag{F.2}$$

Proof. On R, the actual core has between 3 and $\varepsilon_0 M$ edges. The first-moment summation of Appendix B requires only $d \le AM$, independently of the distance from the threshold, so $\Pr(R) = O(M^{-b_k})$; moreover $b_k - (k-2) \ge 1$, so this is $o(\lambda_M)$.

For the intersection, first consider a fixed window. Suppose $d/M = \alpha_c + r/\sqrt{M} + o(M^{-1/2})$ with r fixed, plant a pair of identical uniform k-subsets, and let the remaining edges be independent. The finite planting lemma of Section 5.5 allows the special edges to be mutually correlated, so the precritical score still has the same Gaussian limit; the two repeated edges are themselves legitimate k-subsets, the probability of having no bad edge among the remaining regular edges has a positive limit, and the inclusion–exclusion transfer proceeds as usual. The planted repeated edge makes final failure inevitable, so here only the macroscopic event is transferred: when the score lies below a fixed negative gap, the process stops with $(\alpha_c q_c^k + o(1))M$ edges remaining, which belongs to A; when the score lies above a fixed positive gap, the concentration before stopping and the local freezing rule out early stopping until fewer than $\varepsilon_0 M/2$ edges remain, which does not belong to A. Removing the safety gap gives $Q_r(A) \to \Phi(g_k r)$, where Q_r is the distribution after planting.

Return to (F.1). For any fixed $r_0 < 0$, eventually $d \le \lfloor \alpha_c M + r_0\sqrt{M} \rfloor$. Couple by adding edges using the same sequence of regular edges; the macroscopic event is monotone in the addition of edges, so $\limsup_M Q_M(A) \le \Phi(g_k r_0)$. Letting $r_0 \to -\infty$ gives $Q_M(A) \to 0$. The density of the random-label planting distribution relative to the original distribution is N/λ_M, where N is the number of repeated edge pairs, so

$$\Pr(A \cap D) \le \mathbb{E}[N \mathbf{1}_A] = \lambda_M Q_M(A) = o(\lambda_M). \qquad \square$$

The probability asymptotics of Proposition 7 of the main text hold for any $\varepsilon_0 < \alpha_c q_c^k$, so the macroscopic probability itself is independent of the choice of the cutoff. Proposition 8, Proposition 10, and the conclusions of this appendix conditioned on A all refer to the cutoff chosen in Lemma F.1. When incorporated into the main text, the hypothesis of Proposition 8 is correspondingly changed to "take ε_0 sufficiently small".

F.3 All-positive signs: the two-dimensional initial Doob decomposition

We first work in the configuration model. Let $n_e = kd$ be the total number of endpoints, $\mu = n_e/M$, the endpoints $V_1, \dots, V_{n_e}$ independent uniform, and D_v the occupancy count. Define

$$K = \sum_v (D_v)_2, \qquad Y_K = \frac{K - \mathbb{E}K}{\sqrt{2}\mu\sqrt{M}}, \qquad H = \sum_v h_c(D_v), \tag{F.3}$$

where $(j)_2 = j(j-1)$, $h_c(j) = j(1-a_c)^{j-1}$, and $\mathbb{E}K = n_e(n_e - 1)/M$. The quantity Y_K is the standardization of the occupancy energy, and it is converted back to X_+ in (F.14).

After revealing the first l endpoints, denote the occupancy count by $D_v^{(l)}$, and set $m = n_e - l - 1$. For a function f define $\nu_m^f(j) = \mathbb{E}[\Delta f(j + \mathrm{Bin}(m, 1/M))]$, where $\Delta f(j) = f(j+1) - f(j)$. When the next endpoint V lands, the Doob increment of $\sum_v f(D_v)$ is

$$\Delta_f = \nu_m^f(D_V^{(l)}) - \frac{1}{M} \sum_v \nu_m^f(D_v^{(l)}). \tag{F.4}$$

Taking $f(j) = (j)_2$ gives $\nu_m^f(j) = 2j + 2m/M$, so the Doob increment of the energy is exactly $2(D_V^{(l)} - l/M)$. The total number of endpoints is fixed sample by sample, and here no Poisson family replaces the true initial state.

Lemma F.2 (Mean of the bracket)**.** *Denote the matrix bracket of the initial vector* $(Y_K, (H - \mathbb{E}H)/(\mu_c \sigma_k \sqrt{M}))$ *by* (V_{EE}, V_{EH}, V_{HH})*. Then*

$$\mathbb{E}V_{EE} = (1 - 1/n_e)(1 - 1/M), \qquad \mathbb{E}V_{EH} = \eta_k + O\big((1 + |r_M|)/\sqrt{M}\big), \tag{F.5}$$

and $\mathbb{E}V_{HH}$ *is the same as the initial bracket of Appendix C.*

Proof. Expanding by the collision indicators, $\mathrm{Var}\, K = 2n_e(n_e - 1)(M - 1)/M^2$. For any h, with $B \sim \mathrm{Bin}(n_e - 2, 1/M)$, there is the finite identity

$$\mathrm{Cov}(K, H) = \frac{n_e(n_e - 1)(M - 1)}{M^2} \mathbb{E}\Delta^2 h(B). \tag{F.6}$$

The proof is as follows. Write $K = \sum_{i \neq j} \mathbf{1}_{\{V_i = V_j\}}$. Conditioned on $V_1 = V_2$, the mean of H is $\mathbb{E}h(B + 2) + (M - 1)\mathbb{E}h(B)$; the unconditional mean is $M\mathbb{E}h(B) + 2\mathbb{E}\Delta h(B) + M^{-1}\mathbb{E}\Delta^2 h(B)$. The difference is $(1 - M^{-1})\mathbb{E}\Delta^2 h(B)$, and multiplying by $n_e(n_e - 1)/M$ gives the result.

For $h = h_c$, the Poisson limit gives $\mathbb{E}_{\mathrm{Pois}(\mu)} \Delta^2 h_c(D) = a_c e^{-\mu a_c}(\mu a_c - 2)$. The function $\Delta^2 h_c$ is bounded, the total-variation error between the binomial and the Poisson is $O(M^{-1})$, the parameter μ lies in a compact interval, and the dependence is smooth. By the Doob isometry, $\mathbb{E}V_{EE}$ and $\mathbb{E}V_{EH}$ equal the corresponding variance and covariance divided by the normalizing constants; substituting $\mu = \mu_c + O(|r_M|/\sqrt{M})$ gives (F.5). The limit of the cross term is exactly η_k; it comes from the true occupancy model, with no a priori independence between the dynamic noise and the energy. □

The maximum occupancy count may reach order $\log M$, so the energy increment is unbounded. To obtain concentration, take $L = \lceil \log M \rceil$ and define the truncated energy e_L by $e_L(0) = 0$ and $\Delta e_L(j) = 2\min(j, L)$. For $j \leq L + 1$ we have $e_L(j) = (j)_2$, and $|\Delta e_L| \leq 2L$, $|\Delta^2 e_L| \leq 2$. The true Doob martingale is reconstructed with e_L rather than by simply dropping the large increments of the original martingale.

Lemma F.3 (Super-polynomial concentration of the truncated bracket)**.** *For* $(f, g) \in \{(e_L, e_L), (e_L, h_c), (h_c, h_c)\}$*,*

$$\Pr\Big(|V_{fg}^{(L)} - \mathbb{E}V_{fg}^{(L)}| > CL^2(\log M)^2/\sqrt{M}\Big) \leq 2e^{-c(\log M)^4}, \tag{F.7}$$

and the difference between $\mathbb{E}V_{fg}^{(L)}$ *and the mean of the untruncated bracket, multiplied by any fixed power of* M*, still tends to zero.*

Proof. Changing one input endpoint changes only two occupancy coordinates at each later l, each by 1. For e_L, $\nu_m^{e_L}$ is bounded by $O(L)$, and the per-coordinate change is $O(1)$; for h_c, both are $O(1)$. Hence each $q_l^{fg} = M^{-1} \sum_v \nu_m^f \nu_m^g - \bar{\nu}_m^f \bar{\nu}_m^g$ changes by at most CL^2/M; after normalizing and summing over $O(M)$ steps, the sensitivity of the entire bracket to each input coordinate is at most CL^2/M. McDiarmid's inequality gives (F.7).

The probability that the maximum occupancy count exceeds L is at most $M(eC/L)^L$, which is smaller than any fixed negative power of M. On its complement, K equals the truncated energy, while the absolute values of the two over the whole sample are at most $O(M^2)$. Hence the differences of the center, the variance, and the covariance with H, multiplied by any fixed power of M, still tend to zero. Here only the replacement of the mean is needed; the truncated bracket is not required to equal the original bracket pathwise. □

F.4 Concatenation and the deterministic trace upper bound

After the initial state, the labeled stubs are assigned independently and uniformly to the edges, and the dynamic filtration follows the construction of Appendix C, without revealing the complete grouping in advance. The initial increments of the two-dimensional vector are $(\Delta e_L/(\sqrt{2}\mu\sqrt{M}), \Delta h_c/(\mu_c\sigma_k\sqrt{M}))$, and the dynamic increments are $(0, \text{score martingale increment})$. In the dynamic phase the first coordinate is zero, so it contributes no new EE or EH bracket; this is an identity on the filtration and does not require an independence assumption.

By Lemma F.2, Lemma F.3, and input (C2), except for a super-polynomially small exceptional set, the total matrix bracket $\mathcal{V}$ satisfies

$$\|\mathcal{V} - \Sigma\| \le v_M, \qquad \Sigma = \begin{pmatrix} 1 & \eta_k \\ \eta_k & 1 \end{pmatrix}. \tag{F.8}$$

Σ is positive definite: the initial energy variance is 1, and the score has strictly positive dynamic variance τ_k^2, so $|\eta_k| < 1$.

After truncation the increment bound is $\beta_M = CL/\sqrt{M}$, but over the whole sample the trace bracket is only known to be $O(L^2)$. To perform the exponential tilt, stop the entire vector at the first time the trace bracket exceeds a fixed constant $C_0 > \operatorname{tr}\Sigma + 1$. The bracket increment is a positive semidefinite matrix, so the trace is monotone; the stopping changes the terminal value only when the total trace exceeds C_0, an event of super-polynomially small probability. After stopping, $\operatorname{tr}\mathcal{V} \le C_0 + C\beta_M^2$ holds deterministically. Below, (Y^*, S^*) denotes this stopped vector. On the good set it coincides with the original score and the original energy, up to a super-polynomially small deterministic center term. The subsequent exponential tilt acts only on S^*.

F.5 The two-dimensional rectangle error under a single-coordinate tilt

This section gives the concrete form of the "explicit polynomial rate" of Appendix E of the main text.

Lemma F.4 (Single-coordinate tilt lemma)**.** *Let $\zeta_j = (u_j, v_j)$ be two-dimensional martingale differences with $|\zeta_j| \le \beta$, suppose the trace of the total matrix bracket $\mathcal{V}$ has a deterministic upper bound, and suppose $\Pr(\|\mathcal{V} - \Sigma\| > v) \le q$, where β, v are polynomially small in M and q is super-polynomially small. Write $(U, S) = \sum_j \zeta_j$, let $1 \le t \le C\sqrt{\log M}$, and set the tilted measure $dP_t/dP = e^{-tS}/\mathbb{E}e^{-tS}$. Then the distance ϵ_M between the lower-rectangle CDF of $(U + \eta_k t, S + t)$ under P_t and that of $N(0, \Sigma)$ satisfies*

$$\epsilon_M \le C\Big\{(\beta + v + t\beta + q_Q)^{1/64} + tv + t^2\beta + t^2 v + t^3\beta + q_Q\Big\}, \tag{F.9}$$

where q_Q is super-polynomially small. Substituting $\beta = O(\log M/\sqrt{M})$ and (F.8), the right-hand side is a fixed negative power times a polylogarithm, so $t\epsilon_M \to 0$.

Proof. Step 1: product tilt. Define the measure Q with density $\prod_j e^{-tv_j}/\mathbb{E}(e^{-tv_j} \mid \mathcal{F}_{j-1})$. The deterministic trace upper bound gives $\mathbb{E}(dQ/dP)^2 \le e^{Ct^2}$, so the bad set is still super-polynomially

small under Q; the total-variation distance between the true tilt P_t and Q is $O(t^2v + t^3\beta)$ plus a super-polynomially small quantity.

Step 2: drift and bracket. Write $\mathcal{B}_j = \mathbb{E}(\zeta_j\zeta_j^\top \mid \mathcal{F}_{j-1})$ and $e_2 = (0,1)^\top$. A stepwise Taylor expansion gives

$$\mathbb{E}_Q(\zeta_j \mid \mathcal{F}_{j-1}) = -t\mathcal{B}_j e_2 + \varrho_j, \qquad |\varrho_j| \le Ct^2\beta \operatorname{tr}\mathcal{B}_j, \tag{F.10}$$

and the centered conditional covariance differs from $\mathcal{B}_j$ in operator norm by at most $Ct\beta \operatorname{tr}\mathcal{B}_j$. Let T be the centered vector martingale under Q; then on the good set

$$\|\langle T\rangle - \Sigma\| \le v + Ct\beta, \qquad |(U + \eta_k t, S + t) - T| \le C(tv + t^2\beta).$$

The drift $-\eta_k t$ of the first coordinate comes from the cross entries of $\mathcal{B}_j$ in (F.10); knowing only that the scalar variance is close to 1 does not yield this drift, so the EH bracket must be controlled simultaneously.

Step 3: quantitative two-dimensional rectangle CLT. For a centered two-dimensional martingale T, suppose the increment bound is b, the trace has a deterministic upper bound, and the matrix bracket is within w of Σ with probability at least $1 - q'$. Set $\delta = b + w + q'$. For each unit direction θ, standardize $\theta \cdot T$ by $(\theta^\top\Sigma\theta)^{1/2}$ and stop when the directional bracket first crosses $1 + Cw$; the stopping changes only samples of probability at most q'. A one-step Taylor expansion of the compensated characteristic function gives, direction by direction, for $r = |u| \le \delta^{-1/16}$,

$$|\varphi_T(u) - e^{-u^\top\Sigma u/2}| \le C\{br^3 + b^2r^4 + (w + b^2 + q')r^2 + q'\} \le C\delta(1 + r^4),$$

with the constant uniform in the direction. Here only the characteristic function of the original T is estimated direction by direction; no common stopping vector is constructed.

Add an independent $N(0, h^2I_2)$ to both sides, with $h = \delta^{1/32}$. Fourier inversion gives a supremum bound on the difference of the smoothed densities: the integral over the low-frequency ball $|u| \le \delta^{-1/16}$ is at most $C\delta^{5/8}$, and the high-frequency part is controlled by $e^{-h^2|u|^2/2}$ and is super-polynomially smaller than δ. The second moments of both sides are uniformly bounded; restricting the lower rectangle to $[-R, R]^2$ with $R = \delta^{-1/32}$, the probability outside the region is at most C/R^2, and the integral of the density error inside the region is at most $CR^2\delta^{5/8}$, so the smoothed rectangle error is at most $C\delta^{1/16}$. Finally remove the smoothing: take the coordinate tolerance $a = h\sqrt{C\log(1/\delta)}$; the probability that the noise exceeds the bound is smaller than δ^2; the marginal density of the smoothing Gaussian is uniformly bounded, and the probability of the strip along the rectangle boundary is $O(a)$. Sandwiching the original rectangle between the $\pm a$ translates of the smoothed rectangle gives

$$d_{\mathrm{rect}}(T, N(0,\Sigma)) \le C\delta^{1/64}. \tag{F.11}$$

This conclusion also covers unbounded endpoints, namely the marginal distributions.

Applying (F.11) to the T of Step 2 (with $b = 2\beta$ and $w = v + Ct\beta$), controlling the shift $C(tv + t^2\beta)$ by the anti-concentration of the Gaussian marginals, and adding the total-variation error of Step 1 gives (F.9). □

Remark F.5. *The rate in (F.9) is very weak; it serves only to guarantee $t\epsilon_M \to 0$, and is not used to predict the error at finite M.*

F.6 The macroscopic conditional CDF of the all-positive-sign measurement

Proposition F.6 (Macroscopic conditional law for all-positive signs)**.** *Under (F.1), with the cutoff of Lemma F.1, the Plain model satisfies*

$$\sup_x \left| \Pr(X_+ \le x \mid A) - \frac{\Phi_2(x, -t_0; \eta_k)}{\Phi(-t_0)} \right| \to 0. \tag{F.12}$$

Equivalently, $X_+ + \eta_k t_0$ conditioned on A tends to $N(0, 1 - \eta_k^2)$.

Proof. Inverse tilt. The stopped S^* satisfies the scalar Laplace approximation $\log \mathbb{E}e^{-tS^*} = t^2/2 + o(1)$; take $t = b_M$. For any y, set $\mathcal{K}_M(y) = \mathbb{E}_t[e^{t(S^*+t)}\mathbf{1}_{\{S^*+t\le 0,\ Y^*+\eta_k t\le y\}}]$. Integration by parts in the Lebesgue–Stieltjes sense writes $\mathcal{K}_M$ as an integral of the tilted joint CDF, so Lemma F.4 gives $\sup_y |\mathcal{K}_M(y) - \mathcal{K}_G(y)| \le 2\epsilon_M$, where $\mathcal{K}_G$ corresponds to the target Gaussian. We have $\mathcal{K}_G(\infty) = e^{t^2/2}\Phi(-t) \sim (\sqrt{2\pi}t)^{-1}$. The exact inverse-tilt formula is

$$\Pr(Y^* \le x,\ S^* \le -t) = \mathbb{E}e^{-tS^*}\, e^{-t^2}\, \mathcal{K}_M(x + \eta_k t),$$

Controlling the numerator and denominator simultaneously by the difference between $\mathcal{K}_M$ and $\mathcal{K}_G$, the error of the conditional CDF is at most $O(\epsilon_M/\mathcal{K}_G(\infty)) = O(t\epsilon_M) = o(1)$:

$$\sup_x \left| \Pr(Y^* \le x \mid S^* \le -t) - \frac{\Phi_2(x, -t; \eta_k)}{\Phi(-t)} \right| \to 0.$$

Transfer to the macroscopic event. By input (C3) and the truncation and stopping bad sets of this appendix, $\Pr(A \triangle \{S^* \le -t\}) = o(\Phi(-t))$. This probability bound controls simultaneously the joint sub-probability error of all $\{Y_K \le x\}$, without an energy-weighted bound, so the conditional CDF can be transferred. On the good set Y_K and Y^* differ only by a super-polynomially small amount; the density of the selection Gaussian is uniformly bounded, and, given the score, the conditional density of the energy does not exceed $[2\pi(1-\eta_k^2)]^{-1/2}$, and this remains true after mixing. Replace t by t_0: the two Gaussian-tail conditioning events are nested, and the total-variation distance of the conditional distributions is at most the relative difference of the tail probabilities $O(t_0|t - t_0|) = o(1)$. This yields the conclusion for Y_K in the configuration model.

Fixed planting. Fix l special edges, construct the initial two-dimensional martingale with $k(d-l)$ regular endpoints, and then append the special-edge filtration of Appendix C. Replacing μ by $k(d-l)/M$ changes only the $O_l(M^{-1})$ error in (F.5). When the maximum regular degree does not exceed L, the kl special incidences change K by at most $C_l(L+1)$ and H by C_l; after normalization this is $O_l(L/\sqrt{M})$, and putting it into the perturbation of the energy coordinate and the bandwidth of the macroscopic event, multiplied by t, still tends to zero. Write $G_M(x) = \Phi_2(x, -t_0; \eta_k)/\Phi(-t_0)$; then for each fixed l,

$$\sup_x |\Pr_l(Y_K \le x,\ A)/p_M - G_M(x)| \to 0. \tag{F.13}$$

This formula already contains the relative precision of the probability denominator, which an ordinary finite-planting CLT cannot replace.

Plain transfer. Let N_M^{bad} be the number of bad edges, with $dp_M^{\mathrm{bad}} \to \Lambda_k$. Specifying l bad edges is exactly the allowed special-edge model, so for each fixed l,

$$\sup_x \left| \mathbb{E}[\mathbf{1}_{\{Y_K \le x,\ A\}}(N_M^{\mathrm{bad}})_l]/p_M - \Lambda_k^l G_M(x) \right| \to 0.$$

Apply finite-order Bonferroni upper and lower bounds to the nonnegative indicators. Fix the truncation order and let $M \to \infty$; the finite-term error is uniform in x. The difference between the two bounding sums and $e^{-\Lambda_k}$, multiplied by $G_M(x) \le 1$, then vanishes as the truncation order increases. This gives $\sup_x |\Pr(Y_K \le x, A, N_M^{\mathrm{bad}} = 0)/p_M - e^{-\Lambda_k} G_M(x)| \to 0$, and dividing by $\Pr(A, N_M^{\mathrm{bad}} = 0)/p_M \to e^{-\Lambda_k}$ yields the conditional CDF of the Plain model.

Conversion back to X_+. In the Plain model exactly

$$X_+ = \frac{\sqrt{2}\mu\sqrt{M}}{k(1-k/M)s_M} Y_K + \frac{(k-1)\mu}{k(1-k/M)s_M}, \tag{F.14}$$

that is, $X_+ = (1 + O(M^{-1}))Y_K + O(M^{-1/2})$. The quantity $Y_K + \eta_k t_0$ is tight under A, so the transformation error is $o_P(1)$, and using the density upper bound above gives (F.12). The energy of the all-negative-sign case equals that of the all-positive-sign case. □

F.7 Conditional moments and uniform integrability

Convergence in distribution does not yield convergence of moments after shifting the center. This section controls the tails directly under the macroscopic conditioning.

Proposition F.7 (Macroscopic conditional moments)**.** *Under the hypotheses of Proposition F.6, the moments of every fixed positive integer order and the absolute moments of $W_M^+ = X_+ + \eta_k t_0$ under A converge to the corresponding moments of $N(0, 1-\eta_k^2)$. In particular,*

$$\mathbb{E}[X_+ \mid A] = -\eta_k \ell(t_0) + o(1), \qquad \mathrm{Var}(X_+ \mid A) = 1 - \eta_k^2 + o(1). \tag{F.15}$$

For any fixed sign composition independent of the mapping, $\mathbb{E}[X_s \mid A] = \gamma_M\big[-\eta_k \ell(t_0) + o(1)\big]$, where the $o(1)$ is the same all-positive-sign remainder, hence uniform in the composition, and no convergence of ρ_M is required.

Proof. Quantitative CDF input. We first work in the configuration model. The polynomial rectangle error of Lemma F.4, the factor t of the inverse tilt, the relative bandwidth error $O(t(\delta_M + \epsilon_M))$, and the super-polynomially small bad set divided by p_M, together give

$$d_K\big(\mathcal{L}(Y_K \mid A),\ \mathcal{G}_t\big) \le e_M = O(M^{-c}(\log M)^b), \tag{F.16}$$

where $\mathcal{G}_t$ is the distribution of the first coordinate of a standard two-dimensional Gaussian with correlation η_k, conditioned on the second coordinate being $\le -t$, with $c > 0$ and $b < \infty$ fixed. The Bonferroni bounds have not been used here; when transferring to the Plain model later, only qualitative CDF convergence is needed.

Tilted tail bound over the whole sample. Continue with the product tilt Q of (F.10). Under Q, $T_1 = \sum_j (u_j - \mathbb{E}_Q[u_j \mid \mathcal{F}_{j-1}])$ is a centered martingale. By (F.10), over the whole sample,

$$\Big|\sum_j \mathbb{E}_Q[u_j \mid \mathcal{F}_{j-1}]\Big| \le t \sum_j |(\mathcal{B}_j)_{12}| + Ct^2\beta \sum_j \mathrm{tr}\,\mathcal{B}_j \le C_1 t,$$

Here only $|(\mathcal{B}_j)_{12}| \le \mathrm{tr}\,\mathcal{B}_j/2$, the deterministic trace upper bound, and $t\beta \to 0$ are used; the cross bracket on the bad set is not required to be close to η_k. The increments of T_1 are at most 2β and the sum of the predictable variances is at most C_2, so the Bernstein–Freedman inequality gives $Q(|T_1| > x) \le 2\exp\{-x^2/[2(C_2 + 2\beta x/3)]\}$. Set $W^* = Y^* + \eta_k t$; then $|W^* - T_1| \le C_3 t$, so for $x \ge 2C_3 t$,

$$Q(|W^*| > x) \le 2\exp\{-cx^2/(1+\beta x)\}. \tag{F.17}$$

Unbounded moments without total variation. The density of the true tilt relative to Q is exactly $\prod_j m_j(-t)/\mathbb{E}e^{-tS^*}$, where $m_j(-t) = \mathbb{E}(e^{-tv_j} \mid \mathcal{F}_{j-1})$. The deterministic trace upper bound and $t\beta \to 0$ give $\prod_j m_j \le e^{Ct^2}$; combined with $\mathbb{E}e^{-tS^*} \sim e^{t^2/2}$, this yields the deterministic bound $dP_t/dQ \le e^{C_4 t^2} \le M^{D_0}$ over the whole sample, with D_0 fixed. This crude bound is used together with martingale tail bounds whose order can be raised arbitrarily; multiplying a small total-variation distance by a growing energy maximum does not yield a vanishing error.

Truncated tails. Fix a positive integer q. The actual energy satisfies $|Y_K| \le CM^{3/2}$, and the stopped martingale has at most $O(M)$ steps, each of size $O(\log M/\sqrt{M})$, so $|W^*| \le CM^2$ is a deterministic crude bound. Take $R_M = A_q\sqrt{\log M}$ with A_q large enough that $R_M \ge 2C_3 t$ and that (F.17) gives $Q(|W^*| > R_M) \le 2M^{-H_q}$ with $H_q > D_0 + 2q + 2$. Outside $\mathcal{B}_M$, the event A implies $S^* \le -t + C\delta_M$; for any nonnegative $f(W^*)$, the inverse tilt gives

$$\mathbb{E}[f(W^*); A \setminus \mathcal{B}_M] \le \mathbb{E}e^{-tS^*}e^{-t^2}e^{Ct\delta_M}\mathbb{E}_t f(W^*) \le Cp_M\, t\, M^{D_0}\mathbb{E}_Q f(W^*),$$

where the last step uses $\mathbb{E}e^{-tS^*}e^{-t^2}/p_M = O(t)$. Taking $f = |W^*|^q \mathbf{1}_{\{|W^*|>R_M\}}$ gives $\mathbb{E}[|W^*|^q \mathbf{1}_{\{|W^*|>R_M\}}; A \setminus \mathcal{B}_M]/p_M \le CtM^{D_0+2q-H_q} = o(1)$. The contribution of the bad set is at most $CM^{2q}\Pr(\mathcal{B}_M)/p_M = o(1)$. Hence

$$\mathbb{E}[|W_K|^q \mathbf{1}_{\{|W_K|>R_M\}} \mid A] \to 0, \qquad W_K = Y_K + \eta_k t. \tag{F.18}$$

Since $R_M \to \infty$, (F.18) by itself is not yet uniform integrability.

Between a fixed threshold and R_M. Let Z_0, G be independent standard normals; the shifted-center variable of the selection Gaussian can be written as $W_t^G = \eta_k(Z_0 + t) + \sqrt{1-\eta_k^2}\,G$, conditioned on $Z_0 \le -t$. For $t \ge 1$ and $x \ge 0$,

$$\Pr(-(Z_0 + t) > x \mid Z_0 \le -t) = \frac{\Phi(-(t+x))}{\Phi(-t)} \le e^{-tx - x^2/2},$$

which follows by shifting the integration variable of the numerator by x. Hence the moments of every order of the overshoot tend to zero, and the moments of every order of W_t^G tend to the corresponding moments of $N(0, 1-\eta_k^2)$. For the q-th moment use $\mathrm{clip}(w, -R_M, R_M)^q$, and for the absolute moment use $\min(|w|, R_M)^q$; their total variation is at most $2R_M^q$, and Stieltjes integration by parts with (F.16) gives that the difference of the truncated expectations is at most $2R_M^q e_M = o(1)$. Then (F.18) and the uniform tail control of the Gaussian remove the truncation, giving the convergence of all moments in the configuration model. Taking the conclusion of order $q+1$ gives the uniform integrability

$$\sup_{M \text{ sufficiently large}} \mathbb{E}[|W_K|^q \mathbf{1}_{\{|W_K| > R\}} \mid A] \le C_q / R. \tag{F.19}$$

Plain transfer. The scalar relative Bonferroni already gives $\Pr_{\mathrm{conf}}(A, N_M^{\mathrm{bad}} = 0)/\Pr_{\mathrm{conf}}(A) \to e^{-\Lambda_k} > 0$. For any nonnegative f,

$$\mathbb{E}_{\mathrm{Plain}}[f(Y_K) \mid A] = \frac{\mathbb{E}_{\mathrm{conf}}[f(Y_K)\mathbf{1}_{\{N_M^{\mathrm{bad}}=0\}} \mid A]}{\Pr_{\mathrm{conf}}(N_M^{\mathrm{bad}} = 0 \mid A)} \le C\,\mathbb{E}_{\mathrm{conf}}[f(Y_K) \mid A],$$

so (F.19) carries over directly to the Plain model. This inequality transfers only the tail control; the limiting distribution comes from the Plain CDF transfer of Proposition F.6. Combining the two gives all moment limits in the Plain model. Finally, convert back to X_+ by (F.14): $X_+ + \eta_k t_0 = (1 + O(M^{-1}))(Y_K + \eta_k t) + O(M^{-1/2}) + \eta_k(t_0 - t) + O(t/M)$, and the last several terms tend to zero.

Mean and variance. The first- and second-moment conclusions give $\mathbb{E}[X_+ \mid A] = -\eta_k t_0 + o(1)$ and $\mathrm{Var}(X_+ \mid A) = 1 - \eta_k^2 + o(1)$. The Mills ratio satisfies $\ell(t) - t = O(1/t)$, so the mean can also be written in the form of (F.15). This is only a rewriting; the remainder is still $o(1)$, and it does not identify the coefficient of order $1/t_0$.

Mean of an arbitrary composition. The event A depends only on the unsigned mapping and is invariant under permutation of the element labels, so the exact projection of Eq. (7) of the main text gives $\mathbb{E}[X_s \mid A] = \gamma_M \mathbb{E}[X_+ \mid A]$. This step does not require a sign CLT under a rare event. □

F.8 Arbitrary sign composition: conditional CDF, variance, and endpoints

Proposition F.8 (Macroscopic conditional law for an arbitrary composition)**.** *Under the hypotheses of Proposition F.6, with an arbitrary fixed sign composition, ρ_M need not converge. Set $G_{M,s}(x) = \Phi_2(x, -t_0; \eta_k \rho_M^2)/\Phi(-t_0)$; then*

$$\sup_x |\Pr(X_s \le x \mid A) - G_{M,s}(x)| \to 0, \qquad \mathrm{Var}(X_s \mid A) = 1 - \eta_k^2 \rho_M^4 + o(1). \tag{F.20}$$

The idea of the proof is to avoid redoing a signed tilt for each sign composition. First generate the unsigned Plain hypergraph, then assign the given numbers of positive and negative signs uniformly; this has the same joint law as the original model for (X_s, A), and A selects only the graph. Hence only two things are needed: the required graph properties still hold with probability tending to one under A; and on these fixed good graphs, the sign CLT holds uniformly. The graph properties used in Theorem 2 of the main text hold only to precision

$o_P(1)$, so they cannot be divided directly by the rare $\Pr(A)$. Below, the exceptional probability is strengthened to be super-polynomially small.

We use the overlap-graph notation of Section 5.6: $m_{ij} = |S_i \cap S_j|$, $\Omega = \sum_{i<j} m_{ij}$, $\Omega_2 = \sum_{i<j} m_{ij}^2$, $\ell_i = \sum_{j\neq i} m_{ij}$, $\Omega_\ell = \sum_i \ell_i^2$, and $\bar{\ell} = 2\Omega/d$.

Lemma F.9 (Super-polynomial concentration of the overlap-graph statistics)**.** *There is a deterministic set of graphs $\mathcal{H}_M$ such that $\Pr(\mathcal{G}_M \notin \mathcal{H}_M)$ is super-polynomially small, and all graphs in $\mathcal{H}_M$ satisfy uniformly*

$$\frac{\Omega}{M} = \frac{\mu_c^2}{2} + o(1), \quad \frac{\Omega_2}{M} = \frac{\mu_c^2}{2} + o(1), \quad \frac{1}{M}\sum_i (\ell_i - \bar{\ell})^2 = \mu_c^2 + o(1), \quad \max_i \ell_i \le kL. \tag{F.21}$$

All the $o(1)$ terms can be taken as $O((\log M)^4/\sqrt{M} + |r_M|/\sqrt{M})$. In particular, $\Pr(\mathcal{G}_M \notin \mathcal{H}_M \mid A) \to 0$, and this exceptional probability, multiplied by any fixed power of M and divided by $\Pr(A)$, still tends to zero.

Proof. Let $L = \lceil \log M \rceil$ and $\mathcal{D}_L = \{\max_v D_v \le L\}$. Since $D_v \sim \mathrm{Bin}(d, k/M)$, we have $\Pr(\mathcal{D}_L^c) \le M(eC/L)^L$, which is super-polynomially small.

Lipschitz bounds. For two graphs in $\mathcal{D}_L$, successively replace the edges with different labels by the edges of the other graph. The incidence set of every intermediate graph is contained in the union of the two endpoint graphs, so the vertex degrees do not exceed $2L$. Replacing one edge in such an intermediate graph affects only the $O(L)$ edge labels overlapping the old or the new edge, and each $m_{ij} \le k$, so the changes of Ω and Ω_2 are both $O(L)$. For the replaced edge $\ell_i = O(L)$, so the squared change is $O(L^2)$; for the remaining $O(L)$ affected labels, ℓ_j changes by $O(1)$, so each square changes by $O(L)$. Hence the total change of Ω_ℓ is $O(L^2)$. Thus the three statistics are CL^2-Lipschitz on $\mathcal{D}_L$ with respect to the Hamming distance on the edge labels. Here the global Lipschitz property is proved along a path on the union of two good graphs, without assuming that the one-step path stays inside $\mathcal{D}_L$.

Extension and concentration. For each statistic f take the McShane extension $\widetilde{f}(\mathcal{G}) = \inf_{\mathcal{G}' \in \mathcal{D}_L}\{f(\mathcal{G}') + CL^2 d_H(\mathcal{G}, \mathcal{G}')\}$ and clip it to the deterministic nonnegative polynomial range of f. The extension equals f on the good set and remains CL^2-Lipschitz; the bad-set probability is super-polynomially small, and both f and the clipped extension are at most $O(M^3)$, so $|\mathbb{E}f - \mathbb{E}\widetilde{f}|$ is super-polynomially small. Applying McDiarmid's inequality to the independent edges,

$$\Pr\big(|f - \mathbb{E}f| > C\sqrt{M}(\log M)^4\big) \le \Pr(\mathcal{D}_L^c) + 2e^{-c(\log M)^4}, \qquad f = \Omega, \Omega_2, \Omega_\ell.$$

Deterministic means. For the overlap H of two independent uniform k-subsets, write $h_1 = \mathbb{E}H = k^2/M$ and $h_2 = \mathbb{E}H^2 = k^2/M + k^2(k-1)^2/[M(M-1)]$. Exactly,

$$\mathbb{E}\Omega = \tbinom{d}{2} h_1, \qquad \mathbb{E}\Omega_2 = \tbinom{d}{2} h_2, \qquad \mathbb{E}\Omega_\ell = d\{(d-1)h_2 + (d-1)(d-2)h_1^2\}.$$

The last formula holds because, given S_i, the m_{ij} are independent and their conditional first and second moments do not depend on S_i. Taking the good set as the intersection of $\mathcal{D}_L$ with the three deviation-threshold events gives (F.21); the third relation is computed via $\Omega_\ell - 4\Omega^2/d$, in which the deterministic constants of the two terms cancel. By input (C4), $\Pr(A) \sim p_M$ is no smaller than a fixed negative power divided by a polylogarithm, so the conditional exceptional probability tends to zero. □

Lemma F.10 (Uniform sign CLT on the good graphs)**.** *Take the exact decomposition $X_s = \gamma_M X_+ + V_M$ with $\mathbb{E}_s[V_M \mid \mathcal{G}_M] = 0$. If $\rho_M \to \rho$ and $|\rho| < 1$, then for each fixed u,*

$$\sup_{\mathcal{G} \in \mathcal{H}_M} \Big| \mathbb{E}_s[e^{iuV_M} \mid \mathcal{G}] - e^{-(1-\rho^4)u^2/2} \Big| \to 0. \tag{F.22}$$

Proof. The sign argument of Section 5.6 and Appendix A is an argument over a fixed sequence of graphs; it uses only (F.21), $\Omega = O(M)$, $\Omega_\ell = O(M)$, and the condition that the positive-to-negative ratio stays at a fixed distance from the endpoints. Specifically, the within-group variance bound of the batch flip uses only $\Omega_\ell = O(M)$, and the variance of the flipped subset and the expectations of the internal edges use only Ω_ℓ and Ω; the number of local terms in the main term is $O(M)$, the maximum degree of the dependency graph and the magnitude of the terms are both $O(\log M)$, and the normalized connected cumulants of fixed order $r \geq 3$ are $O_r(M^{1-r/2} \operatorname{polylog} M)$; the variance of the main term is determined by (F.21). Hence for any sequence $\mathcal{G}_M \in \mathcal{H}_M$, $\mathcal{L}_s(V_M \mid \mathcal{G}_M) \Rightarrow N(0, 1-\rho^4)$. If (F.22) failed, one could extract along a subsequence a deterministic graph violating it; that sequence of graphs would still satisfy all the deterministic bounds, contradicting the conclusion above. □

Lemma F.11 (Per-graph conditional variance and the endpoints)**.** *Uniformly over all feasible compositions, on $\mathcal{H}_M$,*

$$\operatorname{Var}_s(V_M \mid \mathcal{G}) = 1 - \rho_M^4 + o(1). \tag{F.23}$$

Proof. One cannot first compute the unconditional $\mathbb{E}V_M^2$ and then divide by the rare $\Pr(A)$, so use instead the exact per-graph variance. Let a_4 be the mean of the product of the signs of four distinct labels. The finite identity gives

$$\operatorname{Var}_s\Big(\sum_{i<j} m_{ij} s_i s_j \;\Big|\; \mathcal{G}\Big) = \Omega_2(1 - 2\gamma_M + a_4) + \Omega_\ell(\gamma_M - a_4) + \Omega^2(a_4 - \gamma_M^2).$$

Set $v = 1-\rho_M^2$ and expand this finite rational expression uniformly over all feasible compositions; this gives

$$v^2\Omega_2 + \rho_M^2 v(\Omega_\ell - 4\Omega^2/d) + O(\Omega_2/d + \Omega_\ell/d + \Omega^2/d^2).$$

On $\mathcal{H}_M$ we have $\Omega, \Omega_2, \Omega_\ell = O(M)$, so the remainder is $O(1)$; multiplying by the normalization factor $4/[k^2(1-k/M)^2 s_M^2] \sim 2/(\mu_c^2 M)$ gives (F.23). This also yields the crude upper bound $C(v + M^{-1})$. When $|\rho_M| \to 1$, (F.23) and the per-graph Chebyshev inequality give $V_M \to 0$, uniformly over the good graphs; this covers any rate of approach to the endpoints. □

Proof of Proposition F.8. Let $U_M = X_+ + \eta_k t_0$. Proposition F.6 gives $U_M \mid A \Rightarrow N(0, 1-\eta_k^2)$. For any subsequence with $\rho_M \to \rho$ and $|\rho| < 1$, first condition on the graph: on the good graphs the inner error tends to zero uniformly by (F.22), on the bad graphs the characteristic function is bounded by one in absolute value, and by Lemma F.9 the probability of the bad graphs under A tends to zero. Hence

$$\mathbb{E}[e^{iaU_M + ibV_M} \mid A] - \mathbb{E}[e^{iaU_M} \mid A]\, e^{-(1-\rho^4)b^2/2} \to 0,$$

that is, U_M and V_M are asymptotically independent under A. When $|\rho| = 1$, use instead the convergence in conditional probability of Lemma F.11.

Keeping the actual shifted center:

$$X_s + \eta_k \rho_M^2 t_0 = \gamma_M U_M + V_M + \eta_k(\rho_M^2 - \gamma_M) t_0,$$

The last term is $O(t_0/d) \to 0$. Here $\rho_M^2 t_0$ is not replaced by $\rho^2 t_0$ in the absence of a rate. Hence $\mathcal{L}(X_s + \eta_k \rho_M^2 t_0 \mid A) \Rightarrow N(0, 1 - \eta_k^2 \rho^4)$. The selection baseline $G_{M,s}$ has the same limit under the same shifted center: it can be written as $c(Z + t_0) + \sqrt{1-c^2}\, G$ conditioned on $Z \leq -t_0$, where $c = \eta_k \rho_M^2$, and the Gaussian overshoot tends to zero. The two distributions tend to the same continuous distribution, so their CDF distance tends to zero. When ρ_M does not converge, take a subsequence on which the CDF distance has a positive lower bound, and by the compactness of $[-1, 1]$ extract a further subsequence on which ρ_M converges, contradicting the conclusion above. This proves the first formula of (F.20).

The variance follows from the law of total variance:

$$\operatorname{Var}(X_s \mid A) = \gamma_M^2 \operatorname{Var}(X_+ \mid A) + \mathbb{E}[\operatorname{Var}_s(V_M \mid \mathcal{G}_M) \mid A].$$

On the good graphs use (F.23); on the bad graphs use the deterministic polynomial upper bound on the energy and the super-polynomial precision of Lemma F.9. Substituting $\operatorname{Var}(X_+ \mid A) \to 1 - \eta_k^2$ from Proposition F.7 and noting $\gamma_M^2 - \rho_M^4 = O(1/d)$ gives the second formula of (F.20). The conditional mean still uses the exact projection and cannot be replaced by the CDF convergence. □

Remark F.12 (Balancing rate)**.** *In the fixed critical window, $\rho_M \to 0$ suffices to recover the standard normal. In the moderate deviation regime, the center $-\eta_k \rho_M^2 t_0$ of the macroscopic layer grows with t_0: if $\rho_M^2 t_0 \to 0$ (which the exactly balanced composition satisfies), then $X_s \mid A \Rightarrow N(0,1)$; if $\rho_M^2 t_0 \to b \in (0,\infty)$, the limit is $N(-\eta_k b, 1)$. This refers to the macroscopic layer. When the weight of the repeated-edge layer in F tends to one, the failure conditional law still recovers normality.*

F.9 Mixing of the two regimes and the weighted mean

Lemma F.1 gives the exact sizes of the parts in the set decomposition $F = A \cup D \cup R$. The planting transfer of Appendix B gives $\sup_x |\Pr(X_s \le x \mid D) - \Phi(x)| \to 0$, which requires only that d/M tends to a finite positive number, and holds equally for $\alpha_M \to \alpha_c$. Hence

$$\Pr(X_s \le x, F) = \Pr(A) \Pr(X_s \le x \mid A) + \lambda_M \Phi(x) + o(\lambda_M),$$

with the error uniform in x. Substituting input (C4) and Proposition F.8 gives the mixture formula of Proposition 8 of the main text. The argument of Section E.2 of Appendix E of the main text is exactly this step; here only the cutoff condition is added.

The $o(1)$ error of the mixture CDF cannot be multiplied directly by the growing macroscopic center, so the conditional mean must be controlled separately on the unnormalized first moment.

Proposition F.13 (Weighted mean)**.** *Under (F.1), take the all-same-sign measurement $Y = s_M X_+$, let N be the number of repeated edge pairs, and set $D = \{N \ge 1\}$. Then*

$$\mathbb{E}[Y; F] = 2\lambda_M - s_M \eta_k \phi(t_0) + o\big(\lambda_M + s_M \phi(t_0)\big). \tag{F.24}$$

Proof. The exact density of the random-label planting gives $\mathbb{E}[YN] = 2\lambda_M$. Set $R = F \setminus (A \cup D)$; per graph there is the identity

$$Y\mathbf{1}_F - Y\mathbf{1}_A - YN = Y\mathbf{1}_R - YN\mathbf{1}_A - Y(N-1)\mathbf{1}_{D\setminus A}. \tag{F.25}$$

Below, the absolute first moments of the three terms on the right are estimated one by one.

Third-kind failure. The moment estimate of the covering event in Section B.5 of Appendix B uses only $d \le AM$, so it holds uniformly on the current moving load: $\mathbb{E}[|Y|; R] \le C\sqrt{M}\, M^{-b_k} = o(\lambda_M)$, since $b_k - (k-2) \ge 1$. Here the core on R has at most $\varepsilon_0 M$ edges, and no fixed subcritical gap is needed.

Multiplicity of the repeated pairs. The joint condition of any two specified repeated pairs involves at most four special labels, and the inner, cross, and outer energy decomposition still gives $\mathbb{E}|Y| \le C\sqrt{M}$; also $\mathbb{E}[N(N-1)] \le \lambda_M^2$, so $\mathbb{E}[|Y|(N-1); D] \le C\sqrt{M}\lambda_M^2 = o(\lambda_M)$.

Intersection of the macroscopic layer and the repeated layer. Let Q be the repeated-planting distribution of the random-label pairs, whose density is exactly N/λ_M. Then

$$\mathbb{E}[|Y|N; A] = \lambda_M \mathbb{E}_Q[|Y|; A] \le \lambda_M \sqrt{\mathbb{E}_Q Y^2} \sqrt{Q(A)} \le C\lambda_M \sqrt{M p_M}.$$

The bound $\mathbb{E}_Q Y^2 = O(M)$ comes from the energy decomposition after repeated planting in Appendix B and the unconditional variance of the outer part. The estimate $Q(A) \sim p_M$ comes from the fixed-planting version of input (C4): treat two identical legitimate k-subsets as finite special edges, then apply the relative Bonferroni transfer to the remaining regular edges. Set $b = s_M t_0/\lambda_M$; from $\sqrt{p}/(1+bp) \le 1/(2\sqrt{b})$,

$$\frac{\lambda_M \sqrt{M p_M}}{\lambda_M + s_M p_M t_0} \le \frac{1}{2}\sqrt{\frac{M\lambda_M}{s_M t_0}} = O\Big(M^{(5/2-k)/2}/\sqrt{t_0}\Big) = o(1).$$

As $t_0 \to \infty$, $p_M \ell(t_0) = \phi(t_0) \approx p_M t_0$, so the intersection term is $o(\lambda_M + s_M \phi(t_0))$, with no need for a refined weighted local response.

Macroscopic main term. By input (C4) and Proposition F.7, $\mathbb{E}[Y; A] = s_M \Pr(A)[-\eta_k \ell(t_0) + o(1)] = -s_M \eta_k \phi(t_0) + o(s_M \phi(t_0))$. The probability needs only relative $o(1)$ precision, since the target remainder is $o(\lambda_M + s_M \phi(t_0))$.

Substituting the three estimates and the main term into (F.25) gives (F.24). □

Combining $\Pr(F) = (p_M + \lambda_M)(1 + o(1))$ with $c_F\binom{d}{2} = \mathbb{E}[Y \mid F]/2$, (F.24) yields the expansion of $c_F\binom{d}{2}$ in Appendix E of the main text, and hence the sign-reversal-layer conclusion of Proposition 10. For any fixed composition, $\mathbb{E}[\widehat{d}_M - d; F]$ equals exactly the all-same-sign expectation multiplied by γ_M, so both the main term and the remainder project by this factor.